

PROJECT · MISSILE DESIGN AND ANALYSIS

Thermodynamic analysis of the engine

Williams F107-WR-402 turbofan modelled in GasTurb 15: cruise design point and parametric study.

AUTHOR	TOOLS	DATE
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NET THRUST	SPECIFIC FUEL CONSUMPTION	OVERALL PRESSURE RATIO	TURBINE ENTRY TEMPERATURE	BYPASS RATIO	PARAMETRIC CASES
2.90 kN	22.83	14.2	1227 K	1.0	1813
Cruise, Mach 0.72	g/(kN·s)	Fan 2.30 · HPC 6.06	Uncooled turbine	Mixed flow	converged in GasTurb

01 Objective and scope

This section reproduces the thermodynamic cycle of the **Williams F107-WR-402** turbofan, the engine selected to power the missile, and evaluates how thrust and fuel consumption respond to its two most influential design parameters.

Published engine data are complemented with justified estimates for the parameters that are not public. With this model the cruise design point is calculated and a map of 49 × 43 combinations of pressure ratio and temperature is evaluated.

ENGINE DATA SHEET

Type	Two-spool, mixed-flow turbofan
LP spool	4-stage fan + booster; 2-stage turbine
HP spool	Centrifugal compressor; 1-stage turbine
Fuel	JP-10, 43.5 MJ/kg

Cruise design point	COMPLETED	Parametric study	COMPLETED	Off-design study	UPCOMING
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02 Method and data provenance

Every result in this summary comes from one of these four steps. Each table and figure states its origin in a **SOURCE** line.

STEP 1 Input data Published engine data (BPR, TET, stages, shaft speed) and own estimates for the rest. Table on page 2.	STEP 2 Design point GasTurb 15 design calculation at Mach 0.72: overall performance, components and stations. Pages 3 and 4.	STEP 3 Parametric study Sweep of TET and pressure ratio in GasTurb, exported to DatosExportados.xlsx. Pages 4 to 6.	STEP 4 Processing The spreadsheet is read with an in-house script; figures and website are generated from the same file.
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03 Input data

Parameters entered in GasTurb for the design calculation. **OFFICIAL** marks a published F107-WR-402 value; **ESTIMATED**, an own assumption with the justification given.

PARAMETER	VALUE	ORIGIN	JUSTIFICATION
FLIGHT CONDITION · CRUISE AT MACH 0.72, LOW ALTITUDE			
Inlet total temperature, T_1	291.5 K	ESTIMATED	Mach 0.72 at low altitude
Inlet total pressure, P_1	116.5 kPa	ESTIMATED	Mach 0.72 at low altitude
Ambient pressure	101.0 kPa	ESTIMATED	Low altitude
Inlet corrected mass flow, $W2Rstd$	8.0 kg/s	ESTIMATED	Set to obtain ≈ 2.9 kN of thrust with BPR = 1
CYCLE			
Bypass ratio (BPR)	1.0	OFFICIAL	—
Burner exit temperature (TET)	1227 K	OFFICIAL	954 °C
Fan pressure ratio (inner and outer)	2.30	ESTIMATED	2 -stage fan + 2 -stage booster
HP compressor pressure ratio	6.06	ESTIMATED	Tuned for an overall pressure ratio ≈ 13.8 (official value)
Combustion efficiency	0.99	ESTIMATED	Standard value
Fuel heating value	43.5 MJ/kg	ESTIMATED	JP-10, high-density fuel
PRESSURE LOSSES (P_{OUT} / P_{IN})			
Air intake	0.97	ESTIMATED	Annular intake in subsonic flight
Combustion chamber	0.96	ESTIMATED	Compact annular combustor, 4 % loss
Ducts (compressor interduct, bypass, turbine interduct, exit)	0.99	ESTIMATED	Short ducts of a compact engine
TURBOMACHINERY			
HPC isentropic efficiency	0.82	ESTIMATED	Small centrifugal compressor
HPT / LPT isentropic efficiency	0.87 / 0.89	ESTIMATED	Uncooled axial turbines with 1 and 2 stages
HP / LP turbine stages	1 / 2	OFFICIAL	—
Turbine cooling air	0 %	OFFICIAL	Uncooled turbine
Nominal LP spool speed	35,500 rpm	OFFICIAL	—
Nominal HP spool speed	60,000 rpm	ESTIMATED	Typical for a small centrifugal compressor
HP and LP spool mechanical efficiency	0.995	ESTIMATED	Standard value for small engines
AIR SYSTEM AND NOZZLE			
Overboard bleed / power offtake	0 kg/s · 0.5 kW	ESTIMATED	No bleed; gearbox alternator only
HP leakage to bypass / to LPT exit	0.5 % / 0.3 %	ESTIMATED	Minimal leakage
Nozzle thrust coefficient	0.985	ESTIMATED	Standard convergent nozzle

SOURCE GasTurb input table published in the portfolio's "Engine analysis" section (step 1, "Input data"), which lists all 62 model parameters.

04 Design point: flow path

At the cruise condition the model delivers **2.90 kN of net thrust** with a specific fuel consumption of **22.83 g/(kN·s)**. Figure 1 follows the core flow: compression raises the total pressure to 1608 kPa and the combustor heats the gas to 1227 K; the turbines extract the energy that drives the HP compressor and the fan, and in the mixer the hot flow joins the bypass stream before expanding through the nozzle.

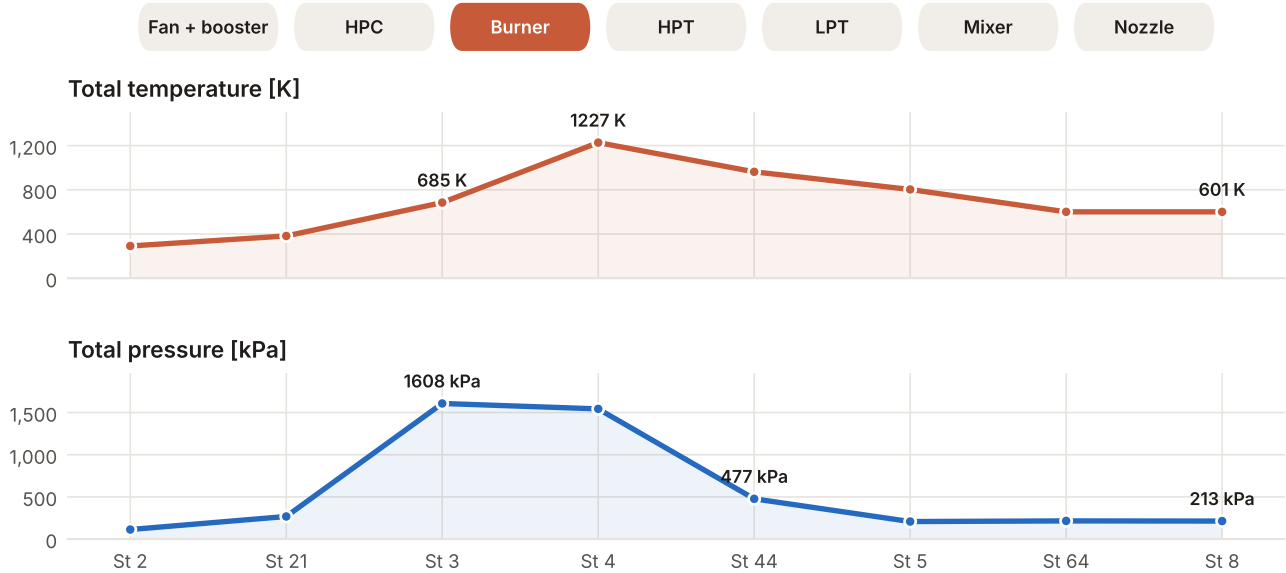


Figure 1 · Total temperature and pressure along the core flow (GasTurb station numbering).

Station properties

STATION	LOCATION	MASS FLOW [KG/S]	TOTAL T [K]	TOTAL P [KPA]	VELOCITY [M/S]	MACH	AREA [M²]
2	Fan inlet	8.871	291.50	113.00	167.0	0.500	0.0444
21	Fan exit, core	4.435	382.28	267.95	191.1	0.500	0.0107
13	Fan exit, bypass	4.435	382.28	251.87	154.2	0.400	0.0136
3	HPC exit	4.435	684.54	1607.54	103.3	0.200	0.0054
4	Burner exit	4.466	1227.00	1543.24	202.6	0.300	0.0053
44	HPT exit	4.466	962.68	477.08	268.6	0.450	0.0106
5	LPT exit	4.466	802.94	207.67	272.8	0.500	0.0205
6	Mixer, core inlet	4.479	802.61	205.60	120.3	0.217	0.0427
16	Mixer, bypass inlet	4.458	383.82	249.35	109.0	0.280	0.0188
64	Mixer exit	8.937	600.51	215.38	193.8	0.404	0.0400
8	Nozzle throat	8.937	600.51	213.23	447.6	1.000	0.0256

SOURCE GasTurb 15 design-point output ("Station Properties"), published in step 2 of the portfolio section. Figure 1 plots the core-flow stations of this table.

05 Design point: performance and cross-checks

The overall quantities reported by GasTurb can be rebuilt from the station table. The last column shows the calculation, so the results can be checked for internal consistency.

QUANTITY	RESULT	HOW IT IS OBTAINED
Net thrust, F_N	2.90 kN	Direct GasTurb output
Fuel flow, W_F	0.0663 kg/s	Direct GasTurb output
Specific fuel consumption (TSFC)	22.83 g/(kN·s)	$W_F / F_N = 66.28 \text{ g/s} \div 2.90 \text{ kN}$
Actual inlet mass flow	8.87 kg/s	$W2Rstd \cdot \delta / \sqrt{\theta} = 8.0 \cdot (113.0 / 101.325) / \sqrt{(291.5 / 288.15)}$
Overall pressure ratio	14.23	$P_3 / P_2 = 1607.5 / 113.0$
Combustor pressure recovery	0.96	$P_4 / P_3 = 1543.2 / 1607.5$
HPT expansion ratio	3.24	$P_4 / P_{44} = 1543.2 / 477.1$
LPT expansion ratio	2.27	$0.99 \cdot P_{44} / P_5 = 0.99 \cdot 477.1 / 207.7$ (turbine interduct loss)
Nozzle pressure ratio	2.11	$P_8 / P_{amb} = 213.2 / 101.0 > 1.85$ (critical): choked nozzle
Nozzle throat area, A_8	0.0262 m²	Direct GasTurb output (station 8)
Core / propulsive efficiency	0.375 / 0.486	Direct GasTurb output

SOURCE GasTurb 15 output ("Overall Performance" and "Station Properties"); own cross-check calculations from those values.

Component performance

COMPONENT	ISENTROPIC η	POLYTROPIC η	P/P
Fan / LPC	0.860	0.875	2.30
HP compressor	0.820	0.858	6.06
Combustor	0.990	—	0.96
HP turbine	0.870	0.853	3.24
LP turbine	0.890	0.879	2.27
Mixer	0.970	—	—

SOURCE GasTurb 15, "Component Efficiencies".

READING THE RESULT

- The nozzle is **choked**: the jet leaves under-expanded, which is usual for convergent nozzles.
- With BPR = 1 the jet is fast and the propulsive efficiency (48.6 %) is moderate: the price of a compact engine that fits inside the fuselage.
- The HP turbine runs **uncooled**, which caps the TET at 1227 K in exchange for a simpler engine.

06 Parametric study

Starting from the design model, GasTurb recalculates the cycle while varying both the burner exit temperature and the HP compressor pressure ratio. All other parameters keep the values on page 2.

ITEM	VALUE
Variable 1 · burner exit temperature (TET)	1062.5 – 1500 K · 43 values (step $\approx$ 10.4 K)
Variable 2 · HP compressor pressure ratio	4 – 10 · 49 values (step 0.125)
Outputs	Specific fuel consumption (TSFC) and net thrust
Cases	2107 set up · 1813 converged · 294 without solution (high PR with low TET)
Check against the design point	PR = 6, TET = 1229 K $\rightarrow$ 22.85 g/(kN·s) and 2.92 kN, vs 22.83 and 2.90 at design

SOURCE data/DatosExportados.xlsx, direct GasTurb export (one block per pressure ratio with "Sp. Fuel Consumption" and "Net Thrust"). Available for download in the portfolio.

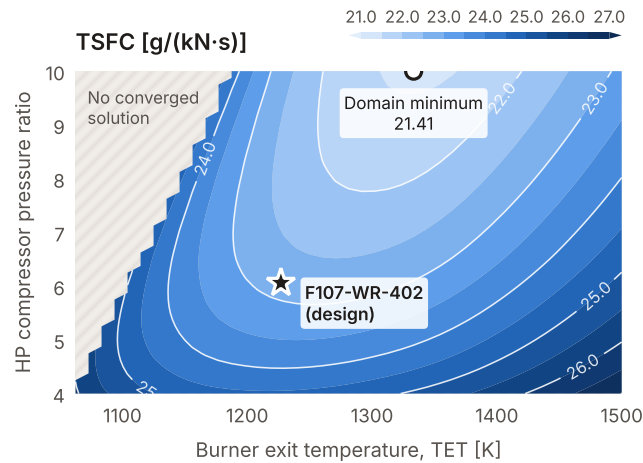


Figure 2 · Specific fuel consumption. Lighter means lower consumption. The hatched area has no converged solution.

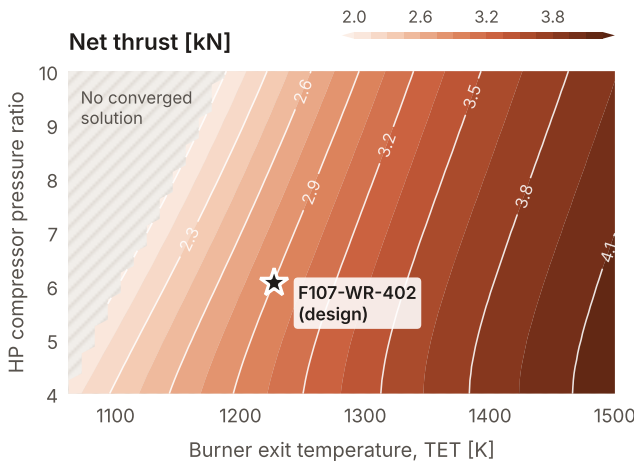


Figure 3 · Net thrust. Darker means more thrust.

Excerpt of the exported data · TSFC [g/(kN·s)] / net thrust [kN]

HPC PR	1104 K	1188 K	1271 K	1333 K	1417 K	1500 K
4	25.25 / 2.36	24.60 / 2.86	24.69 / 3.30	25.08 / 3.59	25.93 / 3.93	27.01 / 4.21
6	24.99 / 2.02	23.16 / 2.65	22.79 / 3.18	23.07 / 3.51	24.00 / 3.86	25.18 / 4.15
8	no solution	23.16 / 2.34	21.99 / 2.97	21.99 / 3.35	22.82 / 3.74	24.01 / 4.05
10	no solution	24.29 / 1.99	21.84 / 2.72	21.41 / 3.17	21.96 / 3.61	23.12 / 3.94

SOURCE data/DatosExportados.xlsx; values without intermediate rounding. The highlighted row contains the lowest consumption of the whole domain (21.41 g/(kN·s) at 1333 K).

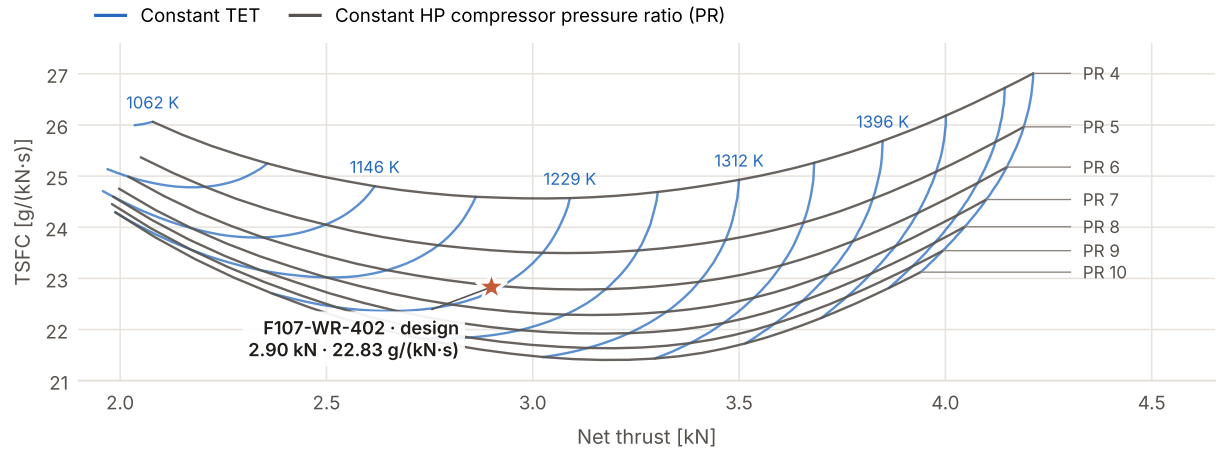


Figure 4 · Thrust-consumption carpet plot built from the same 1813 cases. Each line crossing is a possible design; the star marks the F107-WR-402.

07 Reading the parametric study

1

Temperature drives thrust

At PR = 6, raising the TET from 1104 to 1500 K doubles the thrust (from 2.02 to 4.15 kN). In return, above the optimum temperature consumption rises quickly: at PR = 4 it reaches 27.0 g/(kN·s) at 1500 K.

2

Each pressure ratio has its optimum temperature

The minimum-consumption TET rises with pressure ratio: 1219 K at PR = 4, 1260 K at PR = 6 and 1333 K at PR = 10. Below it the compressor is oversized; above it the jet leaves too fast.

3

At 1229 K, more compression barely pays off

At the design temperature, going from PR = 6 to PR = 8 only lowers consumption by 2 % (22.85 → 22.36) while thrust drops by 9 % (2.92 → 2.67 kN). At PR = 10 consumption rises again.

4

The design is a practical optimum

The global minimum, 21.41 g/(kN·s), is 6.2 % lower than at design, but it needs 106 K more TET (a cooled turbine) and an HP compressor pressure ratio 65 % higher.

SOURCE Figures read from data/DatosExportados.xlsx (figures 2–4 and the excerpt on page 5).

08 Conclusions and next steps

The model is consistent with the published F107-WR-402 data and with itself: **2.90 kN at 22.83 g/(kN·s)** in cruise, and the parametric study reproduces that point within 1 %. The real design gives up 6 % of consumption in exchange for compactness and an uncooled turbine.

Next steps: off-design study (Mach, altitude and power setting) and integration with the aerodynamic analysis, to check that the available thrust matches the missile's cruise drag.

09 Files and sources

data/DatosExportados.xlsx	GasTurb export of the parametric study (1813 cases). Basis of figures 2–4 and of the website.
GasTurb tables in the portfolio	Input data, overall performance, components and stations of the design point (“Engine analysis” section).
Published F107-WR-402 data	Values marked “Official”: BPR, TET, turbine stages, no cooling, LP spool speed and overall pressure ratio.
GasTurb 15	Gas turbine simulation software used for the design point and the parametric study.